# छठी संगोष्ठी: उल्काभ, उल्का, उल्कापिंड: अंतरिक्ष से संदेशवाहक

## 6th Symposium on Meteoroids, Meteors & Meteorites: Messengers from Space

# MetMeSS 2026

**26th to 27th October 2026**

Department of Geology & Geophysics, IIT Kharagpur, in partnership with Physical Research Laboratory, Ahmedabad

**Venue: Kalidas Auditorium, IIT Kharagpur | All times in Indian Standard Time (IST)**

## Day 1: 26th October 2026, Monday

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| **8:30 to 9:30** | **Registration** | |
| **Inaugural ceremony \| 9:30 to 10:30** | | |
| **9:30 to 9:35** | Opening speech: Convenor, MetMeSS-6 (5 min) | **Dr. Sujoy Ghosh**<br>Convenor, MetMeSS-6 |
| **9:35 to 9:40** | **Lighting of the ceremonial lamp and Saraswati Vandana (5 min)** | |
| **9:40 to 9:43** | Address by the Head, Department of Geology & Geophysics (3 min) | **Prof. Sanghamitra Ray**<br>Head, Department of Geology & Geophysics |
| **9:43 to 9:48** | Address by the Chairperson, Kalpana Chawla Space Technology Cell (5 min) | **Prof. Debarati Sen**<br>Chairperson, Kalpana Chawla Space Technology Cell |
| **9:48 to 9:51** | Address by the Dean (Faculty) (3 min) | **Prof. Somesh Kumar**<br>Dean (Faculty), IIT Kharagpur |
| **9:51 to 9:54** | Address by the Dean (Administration) (3 min) | **Prof. Kamal L. Panigrahi**<br>Dean (Administration), IIT Kharagpur |
| **9:54 to 9:58** | Speech by the Convenor, MetMeSS-6 (4 min) | **Prof. Saibal Gupta**<br>Convenor, MetMeSS-6 |
| **9:58 to 10:28** | Inaugural talk by the Chief Guest (30 min) | **Prof. Anil Bhardwaj**<br>Director, Physical Research Laboratory, Ahmedabad |
| **10:30 to 10:40** | **Group photograph** | |
| **High tea: 10:40 to 11:00** | | |
| **Session I: Early Solar System Processes & Organics**<br>*Session Chairs: Ramananda Chakrabarti and Nachiketa Rai* | | |
| **11:00 to 11:15** | **Keynote talk:** From Galactic Matter to Planetesimals: Insights from Meteorites and Sample Returns | **Kuljeet Kaur Marhas**<br>PRL Ahmedabad |
| **11:15 to 11:25** | Beyond the Solar Neighborhood: A New Perspective on Presolar Grain Origins | **Antariksha Mitra**<br>PRL Ahmedabad |
| **11:25 to 11:35** | From Stardust to Supernovae: an Integrated 1D-3D Nucleosynthetic Mixing Framework for Tracing the Origin of Presolar Grains | **Divyadharsini R**<br>PRL Ahmedabad |
| **11:35 to 11:50** | **Keynote talk:** Elemental and Oxygen Isotope Composition of CM Meteorite Constituents: Constraints on Origin of Solar System | **Gopalan Srinivasan**<br>CSSEH Bhopal |
| **11:50 to 12:00** | Characterization of Nanoglobule-like Organics in Mukundpura (CM2): a Coordinated NanoSIMS-hrtem Study | **Shreeya Natrajan**<br>PRL Ahmedabad |
| **12:00 to 12:10** | Nanoscale Carbon and Nitrogen Isotopic Anomalies in Insoluble Organic Matter from the Norton County Aubrite and Parsa Enstatite Chondrite: A NanoSIMS–HRTEM Study | **Neha**<br>PRL Ahmedabad |
| **12:10 to 12:25** | **Keynote talk:** The Construct of Solar System: Deconstructed from High Precision 26Al-26Mg Systematics in Meteorites | **Ritesh Kumar Mishra**<br>AMETEK India, Bengaluru |
| **12:25 to 12:35** | Origin and Evolution of Enigmatic Corundum Inclusions in the Bhuka IAB Iron Meteorite: Insights from $^{26}$Al–$^{26}$Mg and Oxygen Isotope Systematics | **Ankit Prakash Singh**<br>PRL Ahmedabad |
| **12:35 to 12:50** | **Invited talk:** Stable isotope variations of major elements in planet forming materials | **Yogita Kadlag**<br>PRL Ahmedabad |
| **12:50 to 13:00** | C, H, and N isotopic studies of C2 and C1-type clasts from a polymict ureilite breccia. | **Subhasmita Swain**<br>NISER Bhubaneswar |
| **13:00 to 13:10** | Unique olivine-rich volatile-rich C1 clasts in two polymict ureilites | **Subhasmita Swain**<br>NISER Bhubaneswar |
| **13:10 to 13:20** | Oxygen Isotope Concentration of Carbonate and Olivine Grains within C1 and CM-like Clasts in Achondritic Breccia Meteorites. | **Neetha Thomas**<br>IISER Berhampur |
| **Lunch: 13:20 to 14:40** | | |

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| **Session II: Asteroids, Comets, Micrometeorites & Returned Samples**<br>*Session Chairs: Sujoy Ghosh and Kuljeet Kaur Marhas* | | |
| **14:40 to 14:55** | **Keynote talk:** Meteorite Mineralogy and Indian Meteorite Fall | **Dipak Kumar Panda**<br>PRL Ahmedabad |
| **14:55 to 15:05** | Deciphering Pre- and Post-Accretionary Processing of Organic Matter in Asteroidal Materials | **Rahul Kumar**<br>IIT (ISM) Dhanbad |
| **15:05 to 15:15** | Phosphate mineralogy and fluid-assisted alteration in the Khalwat-Nimgaon L5 ordinary chondrite: Implications for S-Type asteroid parent bodies | **Garima Arora**<br>PRL Ahmedabad |
| **15:15 to 15:30** | **Keynote talk:** Cosmic Ray Exposure Ages of Meteorites and Surface Exposure Systematics of Planetary Objects | **Ramakant R. Mahajan**<br>PRL Ahmedabad |
| **15:30 to 15:40** | Neutron Signatures in Ordinary Chondrites | **Satvika Jaiswal**<br>PRL Ahmedabad |
| **15:40 to 15:50** | Thermal Overprinting by Fusion Crust Controls Back-transformation of High-pressure Phases in the Katol L6 Chondrite | **Kishan Tiwari**<br>IIT Kharagpur |
| **Tea break: 15:50 to 16:05** | | |
| **Session II: Asteroids, Comets, Micrometeorites & Returned Samples (contd.)** | | |
| **16:05 to 16:20** | **Keynote talk:** Shock Features in Ordinary Chondrites: Observations from the Geological Survey of India Archive | **Arindam Dutta**<br>GSI Kolkata |
| **16:20 to 16:30** | Olivine-ringwoodite transformation mechanism in Bori L6 chondrite. Implication for slab rheology | **Rajesh Kumar Behera**<br>IIT Kharagpur |
| **16:30 to 16:40** | Interpreting the Entry Conditions for the Development of Different Textures in Micrometeorites | **Vijay Pratap Singh**<br>CSIR-NIO Goa |
| **16:40 to 16:50** | Evaluation Of Highly Oblique Impact Crater Near Luobusa Ophiolite Using, Numerical Models, Ultra High-Pressure Minerals and Tektites | **Mahesh Patil**<br>SSRI Ahmedabad |
| **Session III: Differentiated Bodies: Asteroids, Moon, Mars**<br>*Session Chairs: Gopalan Srinivasan and Yogita Kadlag* | | |
| **16:50 to 17:05** | **Keynote talk:** End Member Models in Planetesimal Formation: Constraints from Differentiated Meteorites | **Nachiketa Rai**<br>IIT Roorkee |
| **17:05 to 17:20** | **Keynote talk:** Non-traditional Isotope Geochemistry of Planetary Materials | **Ramananda Chakrabarti**<br>IISc Bengaluru |
| **17:20 to 17:35** | Audience questions and Session III discussion | **Session chairs** |
| **Break and cultural programme set-up: 17:35 to 18:00** | | |
| **Cultural programme: 18:00 to 20:00**<br>*Kalidas Auditorium* | | |

## Day 2: 27th October 2026, Tuesday

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| **Session III: Differentiated Bodies: Asteroids, Moon, Mars (contd.)** | | |
| **9:00 to 9:10** | Petrography of the Polymict Ureilite Northwest Africa (NWA) 15112: Clast Diversity and Implications for Reduction and Impact Processing | **Aishanya Sharma**<br>IIT Roorkee |
| **9:10 to 9:20** | Late homogenization of mantle sources recorded by 182W isotope signatures from the Singhbhum Craton | **Arathy Ravindran**<br>IISER Thiruvananthapuram |
| **9:20 to 9:35** | **Keynote talk:** Diversity of Lunar Feldspathic Highland Crust: the View After CHANDRAYAAN-3 | **Dwijesh Ray**<br>PRL Ahmedabad |
| **9:35 to 9:45** | How Did Some Basalts on the Moon Become So Rich in Titanium? | **Himela Moitra**<br>IIT Kharagpur |
| **9:45 to 9:55** | Identification and Characterization of Previously Unidentified Events from Apollo Passive Seismic Data | **Omkar Kumar Gupta**<br>PRL Ahmedabad |
| **Session IV: Space Weathering, Meteors, Atmosphere & Ionosphere**<br>*Session Chairs: Satadru Bhattacharya and Bhalamurugan Sivaraman* | | |
| **9:55 to 10:10** | **Keynote talk:** The PAH problem in the ISM, Bottom-up or Top-down | **Bhalamurugan Sivaraman**<br>PRL Ahmedabad |
| **10:10 to 10:20** | Mid-infrared Spectroscopic investigation of a sample of asteroids | **V. Venkataraman**<br>SPL VSSC Thiruvananthapuram |
| **10:20 to 10:30** | Pulsed Laser Irradiation of Three CM Chondrites: Micrometeorite related Space Weathering on Organic-rich Asteroids | **Swarna Prava Das**<br>NISER Bhubaneswar |
| **10:30 to 10:40** | Solar- and Cosmic Ray Tracks in Regolith Breccia Meteorites | **Neetha Thomas**<br>IISER Berhampur |
| **Tea break: 10:40 to 11:00** | | |
| **Session IV: Space Weathering, Meteors, Atmosphere & Ionosphere (contd.)** | | |
| **11:00 to 11:10** | Stony meteorites at hypervelocities in Venusian atmosphere, An experimental approach | **Arijit Roy**<br>PRL Ahmedabad & IIT Kharagpur |
| **Session V: Planetary Analogues**<br>*Session Chairs: Saibal Gupta and Goutam Dev Mukherjee* | | |
| **11:10 to 11:20** | Role of Fluid into the Origin of Ancient LIP Rhyolite and Large Granitoid: Implications for Martian Silicic Rocks | **Sarajit Sensarma**<br>University of Lucknow |
| **11:20 to 11:30** | Combined Radiogenic K–CA and RB–SR Isotopic Age Constraints of the Dhala Impact Structure, India | **Gaurav Singh Papola**<br>IISc Bengaluru |
| **11:30 to 11:45** | **Keynote talk:** Significance of Terrestrial Analogues for Lunar Exploration: Insights from the Sittampundi Anorthosite Complex, India | **Rajesh V. J.**<br>IIST Thiruvananthapuram |
| **11:45 to 11:55** | Evaporative Polygons: An Indication of Transient Brine Activity on Mars | **Angana Samanta**<br>IIT Bombay |
| **11:55 to 12:05** | 3D printing of space bricks and radiation shield using planetary analogues | **Thirukumaran Venugopal**<br>Chikkaiah Govt. College, Erode |
| **12:05 to 12:20** | **Invited talk:** Raman Spectroscopy of Ilmenite-Bearing Lunar Mantle Analogues at High Pressure | **Goutam Dev Mukherjee**<br>IISER Kolkata |
| **12:20 to 12:35** | **Invited talk:** Laboratory Studies for Planetary Science: Developing a New Experimental Facility at SINP, Kolkata, India | **Sankhabrata Chandra**<br>SINP Kolkata |
| **Lunch: 12:35 to 14:00** | | |
| **Session V: Planetary Analogues (contd.)** | | |
| **14:00 to 14:10** | Extremophile microbiomes hosted in planetary analogue sites across the Indian subcontinent | **Diptangshu Dey**<br>IIT Kanpur |
| **14:10 to 14:20** | Biosignature Preservation in Travertine Deposits from Tapovan Hotspring, Uttarakhand: Implications for Mars | **Pallabi Das**<br>PRL Ahmedabad |
| **Session VI: Planetary Remote Sensing & Instrumentation**<br>*Session Chairs: Dwijesh Ray and Rajesh V. J.* | | |
| **14:20 to 14:45** | **Keynote talk:** Vis-IR Spectroscopy & Planetary Remote Sensing | **Satadru Bhattacharya**<br>SAC ISRO Ahmedabad |
| **14:45 to 14:55** | MRO-HiRISE observations of aeolian characteristics at the Arkhangelsky crater on Mars | **Anirban Mandal**<br>NIT Rourkela |
| **14:55 to 15:05** | Simulation of neutron and gamma spectra of Mars: Constraints on thickness of CO2 frost layer | **Debabrata Banerjee**<br>PRL Ahmedabad |
| **15:05 to 15:20** | **Invited talk:** Chandrayaan-3 provides intriguing insights into thermophysics and geo-technical environment at lunar high-latitudes | **Karanam Durga Prasad**<br>PRL Ahmedabad |

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| **15:20 to 15:30** | 3D Thermophysical Modelling of Martian Surface and Subsurface Temperatures: Validation Against In-Situ Observations | **Karanam Durga Prasad**<br>PRL Ahmedabad |
| **15:30 to 15:40** | Design and Development of a Miniaturized Neutral and Ion Mass Spectrometer (NIMS) for Future Planetary Missions | **Sanjeev Kumar Mishra**<br>PRL Ahmedabad |
| **15:40 to 15:55** | **Invited talk:** Chemical Fingerprints of Planetary Evolution | **Liton Majumdar**<br>NISER Bhubaneswar |
| **15:55 to 16:50** | **Poster session with tea** | |
| **Special keynote talks \| Venus and planetary volcanism** | | |
| **16:50 to 17:10** | **Keynote talk:** Radiating Mafic Dyke Swarms: How Far Can They Go? Insights from Venus | **Hafida El Bilali**<br>Carleton University, Ottawa |
| **17:10 to 17:30** | **Keynote talk:** Applying a Mantle Plume-Large Igneous Province-Mafic Dyke Swarm (MP-LIP-DS) Paradigm to Geological Mapping and Interpretation on Venus | **Richard Ernst**<br>Carleton University, Ottawa (online presentation) |
| **Concluding session \| 17:30 to 18:00** | | |
| **17:30 to 17:45** | **Young Researcher Award presentation** | |
| **17:45 to 17:55** | **Discussion and feedback** | |
| **17:55 to 18:00** | **Vote of thanks** | |
| **Gala dinner with families: 19:30**<br>*Gargi-Maitree Foyer* | | |

# Poster Presentations

Poster viewing: Tuesday, 27 October 2026, 15:55 to 16:50, with tea. Posters must be smaller than 1 m x 1 m, rectangular, preferably 4:3. Most keynote and invited talks are 15 minutes and contributed talks 10 minutes, including questions; the two Venus keynote talks on 27 October are 20 minutes each.

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| **Session I: Early Solar System Processes & Organics** | | |
| **P01** | High resolution spectroscopy and laboratory astrochemistry @IIT Kharagpur | **Arijit Roy**<br>Dept. of Chemistry, IIT Kharagpur |
| **P02** | ℓ-type Doubling in HC2n+1N and HC2n+1P Linear Molecules of Astrochemical Interest | **Ratul Paul**<br>IIT Kharagpur |
| **P03** | Raman Spectroscopic Constraints on the Formation Conditions of Presolar Silicon Carbide Grains from the Murchison Meteorite | **Aadhya Tripathi**<br>IISER Mohali |
| **P04** | Radiative Transfer Modeling of Molecular Line Profiles in the Hot Corino Proto-binary Ngc 1333 IRAS4A | **Bhalamurugan Sivaraman**<br>PRL Ahmedabad |
| **P05** | Mineralogical, Chemical and Shock-Metamorphic Characterization of Chondrule Populations in the Dergaon H5 Chondrite | **Kalpana Singh Verma**<br>IIT Roorkee |
| **Session II: Asteroids, Comets, Micrometeorites & Returned Samples** | | |
| **P06** | Textural and Environmental Discrimination of Extraterrestrial and Terrestrial Spherules: Evidence from Central Indian Ocean Basin Sediments and Antarctic Cryoconite Sediments. | **Krishnaveni D**<br>CSIR-NIO Goa |
| **P07** | Noble Gas Isotope Systematics of H Group Ordinary Chondrites | **Avadh Kumar**<br>PRL Ahmedabad |
| **P08** | Spectral Signatures of Simulated Asteroid Regoliths: Linking Physical Properties to VNIR–SWIR Characteristics | **Sritama Das**<br>IIT Kharagpur |
| **P09** | OH, Organics, and $CO_2$ in the Outer Solar System: A JWST/NIRSpec Spectral Investigation. | **Arnab Chatterjee**<br>IIT Kharagpur |
| **Session III: Differentiated Bodies: Asteroids, Moon, Mars** | | |
| **P10** | Late-stage Silicic Differentiation on Mars: Evidence from Si–Na–K-rich Mesostasis and Inclusions in the Shergotty Shergottite | **Manoj Jat**<br>PRL Ahmedabad |
| **P11** | Reconstructing the Geological Past of Mars: Insights from Olivine Alteration in the Nakhlite Yamato-593 | **Aditya Das**<br>PRL Ahmedabad |
| **P12** | Textural Study of the Martian Meteorite - Shergotty | **Mahima Sharma**<br>IIT Roorkee |
| **P13** | Petrographic and Mineralogical Insights into Winona and NWA 14344: Implications for Winonaite Petrogenesis | **Nishant Pandey**<br>IIT Roorkee |
| **P14** | Probing the origin of Earth's apparent non-chondritic Mg isotope signature | **Soumyadip Haldar**<br>IISER Kolkata |
| **P15** | Geochemical Insights into Basalt Weathering on Mars: a Case Study of Lyot Crater | **Sneha Bhowmik**<br>IIT Kharagpur |
| **P16** | Spectral Anomalies and Possible Iron-meteorite Origins of FE-RICH Float Rocks at Jezero Crater, Mars: a Mastcam-z and Supercam Investigation. | **Subham Sarkar**<br>IISc Bengaluru |
| **Session IV: Space Weathering, Meteors, Atmosphere & Ionosphere** | | |
| **P17** | The Largest Peptide-like Molecule in the ISM- Propionamide in Astrochemical Icy Conditions | **Bhalamurugan Sivaraman**<br>PRL Ahmedabad |
| **Session V: Planetary Analogues** | | |
| **P18** | Geochemical Studies of Impact-melt Rocks from the Lonar Crater | **Anagha S M**<br>PRL Ahmedabad |
| **P19** | Impact-melt spherules at Lonar crater, India, and Jezero crater, Mars: A Terrestrial Analogue perspective | **Himangshu Deka**<br>PRL Ahmedabad |
| **P20** | Mineralogical signatures of basaltic laterites from India: Implications for Fe-rich alteration history on Mars | **Aindrila Mukherjee**<br>PRL Ahmedabad |
| **P21** | Shock Driven Chemical Processing of Lunar Analogues | **Roshan Nath**<br>GITAM Bengaluru |
| **Session VI: Planetary Remote Sensing & Instrumentation** | | |
| **P22** | Studies on Interannual Variability of Argon in the Martian Atmosphere | **Shipra**<br>PRL Ahmedabad |
| **P23** | Deep Learning Classification of Martian Impact Crater Degradation from CTX Imagery | **Saksham Badal**<br>IIT (ISM) Dhanbad |
| **P24** | Can the Experimental Maturity Sequence Help to Locate Relatively Recent Brine Activity on Mars? | **Adrita Mukherjee**<br>IIT Kharagpur |
| **P25** | Deep-Learning-Based Semantic Segmentation of Lunar Swirls | **Subhangini Soni**<br>PRL Ahmedabad |

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| **P26** | Integrated Assessment of Safe Landing Zones and Traverse Planning at the Lunar South Pole | **Suyash Sharma**<br>IIST Thiruvananthapuram |